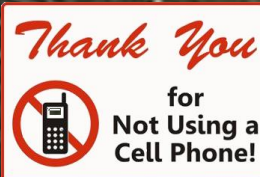


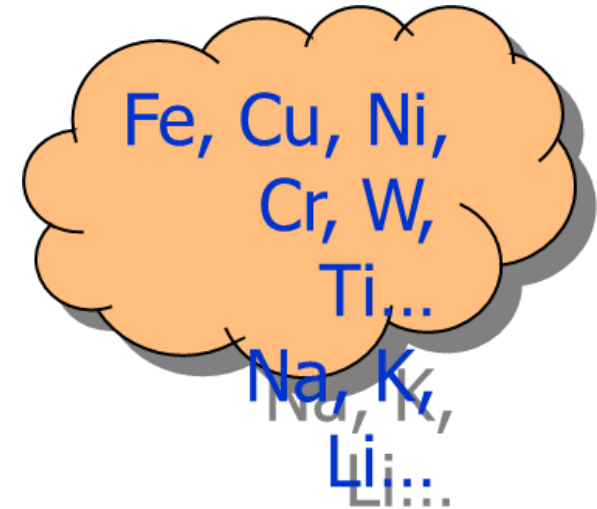
Contamination Control in EKL

December 2025 - Standard EKL version



Outline

- I. What is contamination?
- II. The effects of contamination on semiconductors and MEMS devices.
- III. Contamination prevention in EKL.
- IV. Contamination ethics.



What is contamination?

- **Contamination** can be defined as any process, condition, material or effect that can degrade a device.
- **Contaminants** are any impurities or undesired materials that degrade device performance, quality, yield or affect manufacturability.
- **Cross contamination** is a process by which contaminants are unintentionally transferred or spread from one object to another.

It is important to know the tolerable threshold values of contaminations. Impurity concentrations below the threshold values have no impact on the product performance.

Thresholds vary with product type: Pharmaceutical, food, semiconductors (Bipolar/CMOS/MEMS).

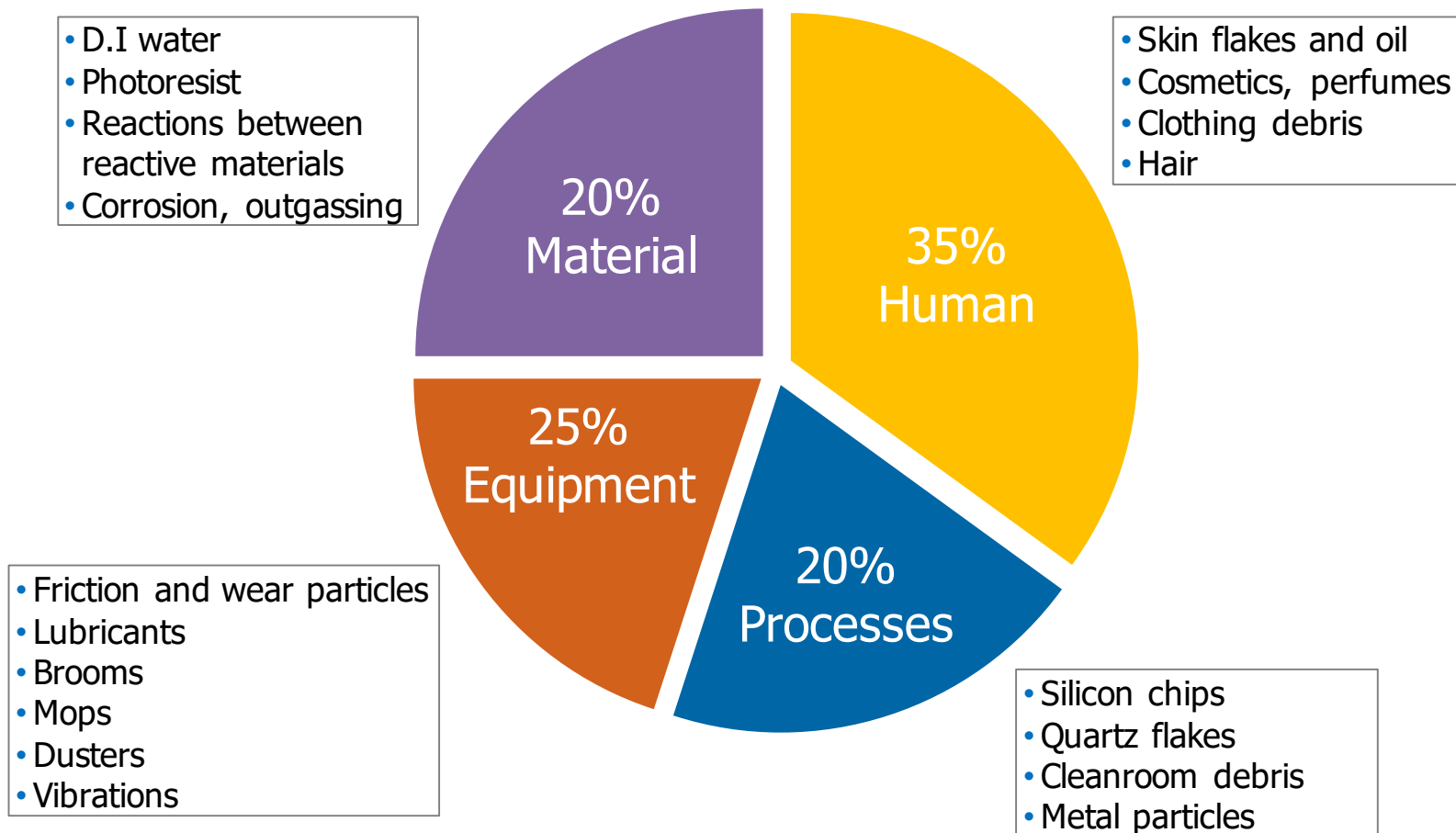
Contamination issues

- Semiconductor manufacturing is a **contamination sensitive** industry. Therefore manufacturing is performed in cleanrooms.



- Contamination is responsible for **75%** of the yield loss in integrated circuit fabrication.

Contamination sources



Transport of contamination

Contaminants can appear at the **surface** of a silicon wafer, or within the **bulk** of a silicon wafer.

Transport mechanisms:

- Brownian movement and **convection**.
- Molecular, chemical **diffusion**.
- Electromagnetic **diffusion**.

Contamination classification

- **Particle contamination**

A particle is a stable or quasi stable agglomeration of molecules with a small feature size: from 2 nm to 1 mm. Biological contaminants as bacteria and viruses are treated as particles.

- **Molecular contamination**

A molecular contamination is caused by any contaminant that is not classified as a particle, such as a single molecule, a small collection of molecules, or a mono-molecular film.

Types:

- Ionic contamination (liquid or solid)
- Airborne Molecular Contaminants (AMC's)

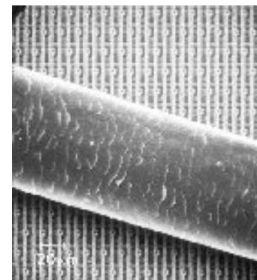
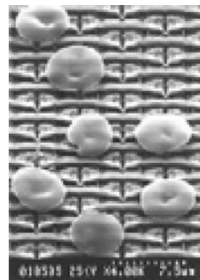
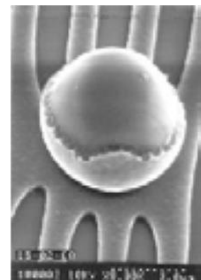
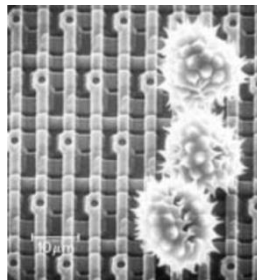
Molecular contamination types

Ionic contamination (liquid or solid)

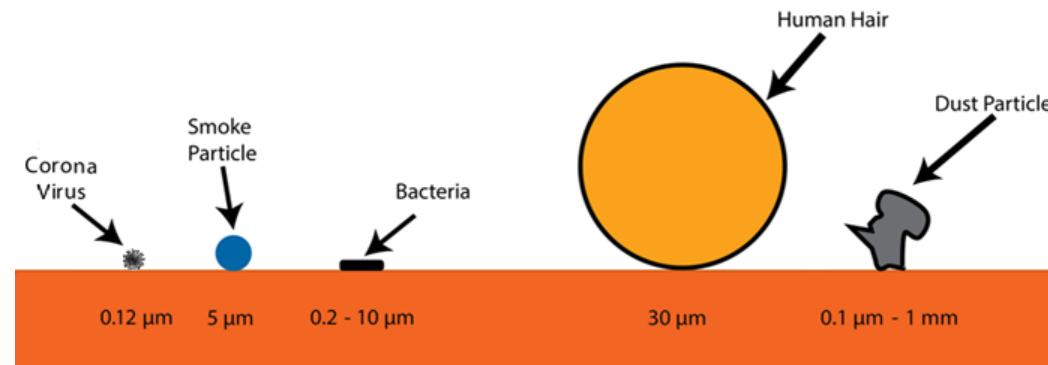
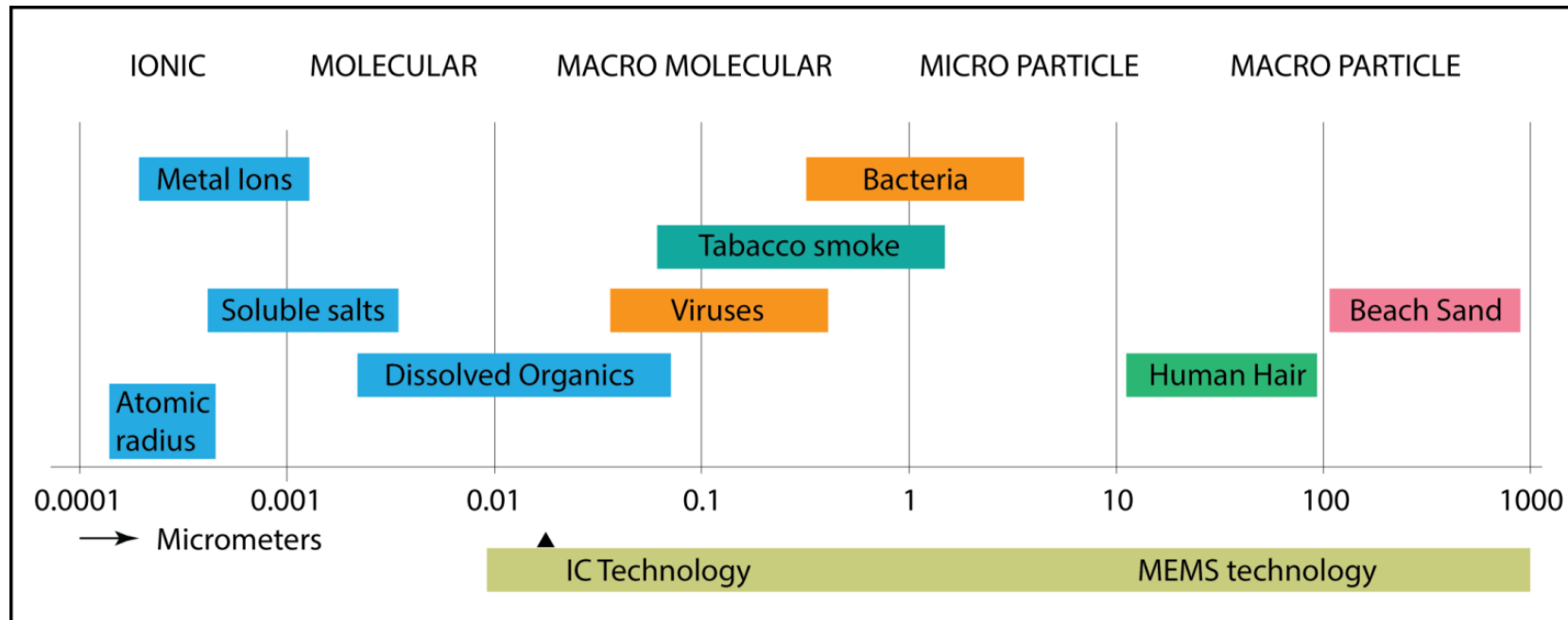
Undesired alkaline or metal atoms deposited (as particles or in ionic form) on the Si surface during device processing, as a result of an error during the process.

Airborne Molecular Contaminants (AMC's)

Airborne Molecular Contamination (AMC) is a chemical contamination in the form of vapours or aerosols, that has a detrimental effect on a product or a process.



Contamination size



Particle sources

- **Cleanroom environment**

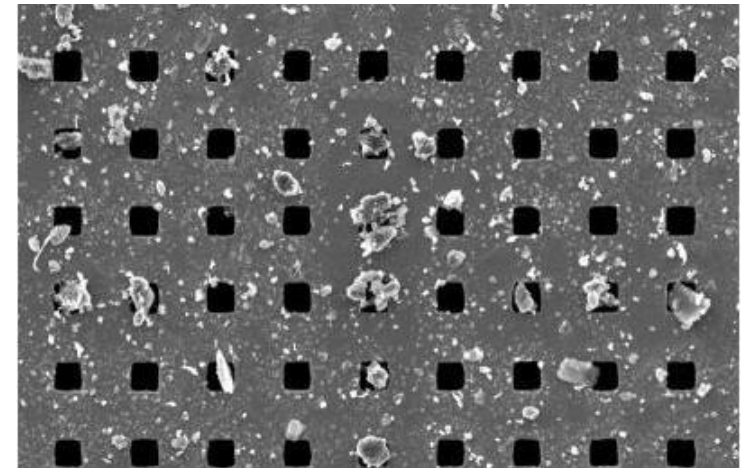
(including the person himself/herself)

- **Equipment**

- Wafer handling and transport (moving platforms, chucks, robot arms, etc.)
- Pumping and venting, valve operations, etc.

- **Wafer processing**

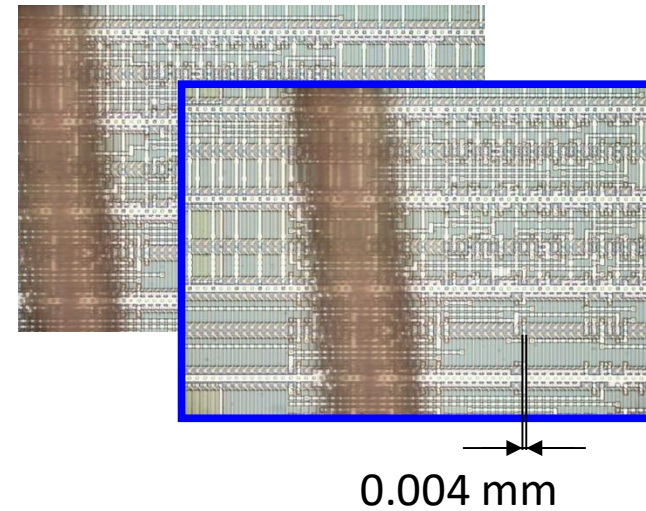
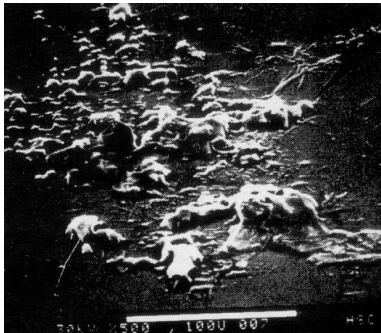
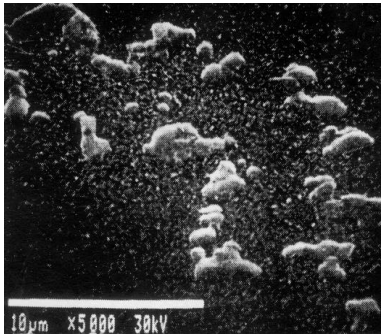
- PECVD
- Plasma etching
- Metal sputtering
- Dicing
- CMP
-



Every wafer processing step is a potential source of contamination.

Particle sources

Cleanroom environment, including people:



Effects of particles

Particle **position** and **dimension** are important:

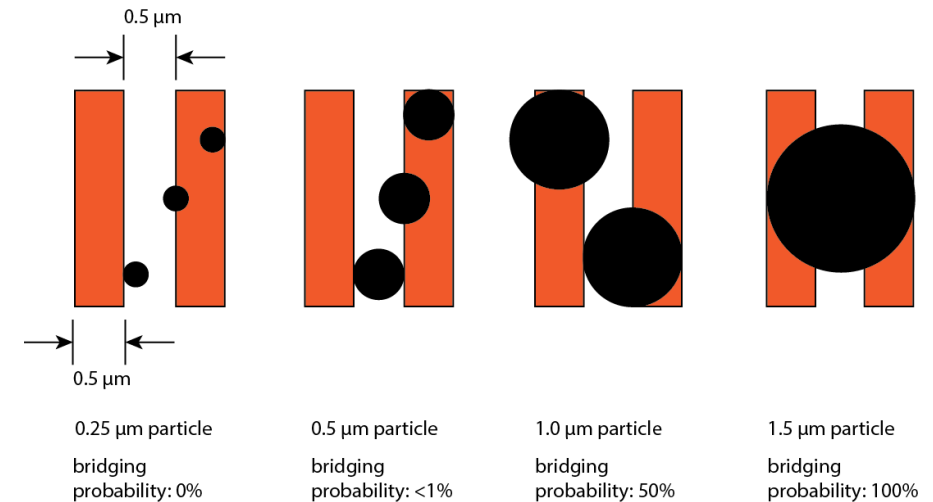
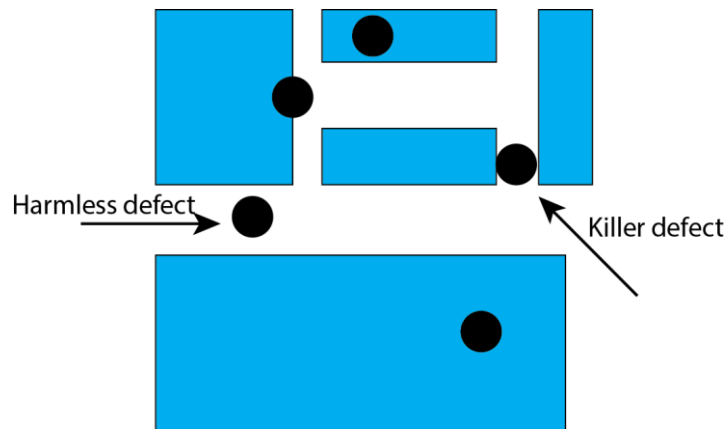
- Particles present in a critical part of the device destroy it's functioning:

→ **killer defects**

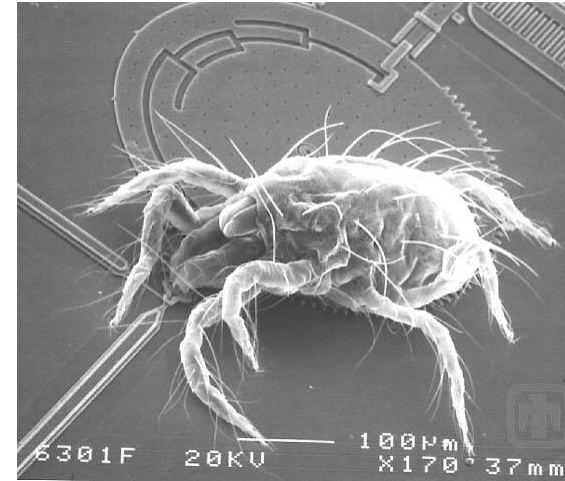
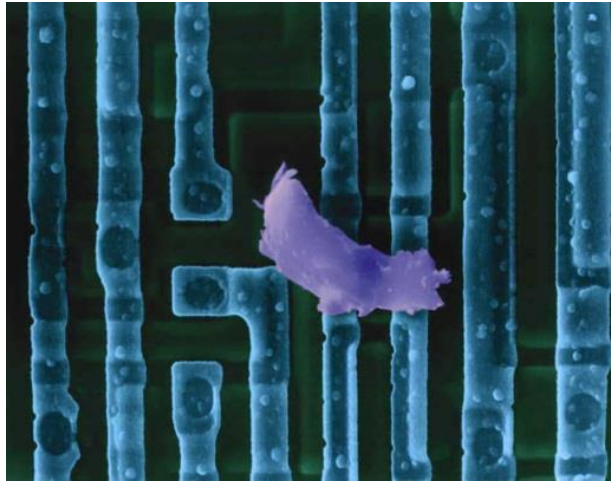


- Particles present in a less sensitive area:

→ **harmless defects**



Particle size



Particle size must be 10 times smaller than the minimum feature size.
(e.g. a $0.30\text{ }\mu\text{m}$ feature size device is vulnerable to $0.03\text{ }\mu\text{m}$ diameter particles.)

Ionic contamination sources

- Equipment
 - Wafer handling and transport (substrate holders, tweezers, ...)
 - Pumping and venting, valve operations
 - Vacuum chambers (physical/chemical reactions)
- Chemicals
- Metals
 - Sputtered or evaporated
- Dopants
 - B, Al, Ga, In, P, As, Sb
- Wafer processing
 - Plasma etching of metals, implantation



10 mg of Fe is sufficient to contaminate 10 ktons of Si to the level of 10^{11} atoms/cm³.

Ionic contamination effects

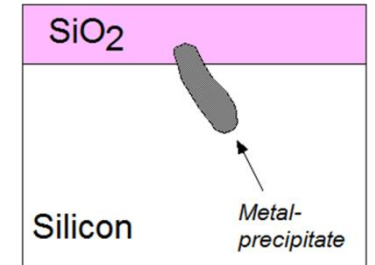
Change of electrical device characteristics and reliability.

Major problems:

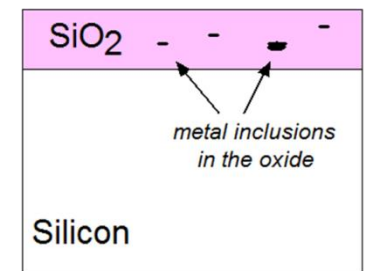
- Oxide reliability
- Dielectric breakdown of gate oxide
- V_T shift
- Low carrier lifetimes
- Junction leakage
- Bonding pad corrosion

Mobile Ionic Contaminants (MIC) must be $< 10^9$ atoms/cm².

Local thinning of the oxide



Trap-assisted tunneling



Sodium is the most common MIC, especially in MOS devices
→ look for low-sodium-grade chemicals

Airborne contaminants sources

- General environment
- Building and cleanroom construction materials
(Boron and Phosphorus from HEPA filters)
- Gasses
(SO_x, NO_x, CO, CF compounds, NH₃, ...)
- Process chemicals
- Operating personnel

Airborne contaminants effects

- Change the wetting characteristics of etchants $\Rightarrow$ incomplete etching.
- Polysilicon nucleation.
- Uncontrolled native oxide formation.
- Film adhesion problems.
- Harmful to processes that require delicate chemical reactions, e.g. the exposure of photoresist in the patterning operations.
- Shifting etch rates.
- Unwanted dopants that shift electrical device parameters.

General effect of contamination

Semiconductor devices

Device processing **yield**

- Contaminants may change the dimensions of device parts, surface cleanliness, give pitted layers, ...
- They can reduce the overall yield during various quality checks.

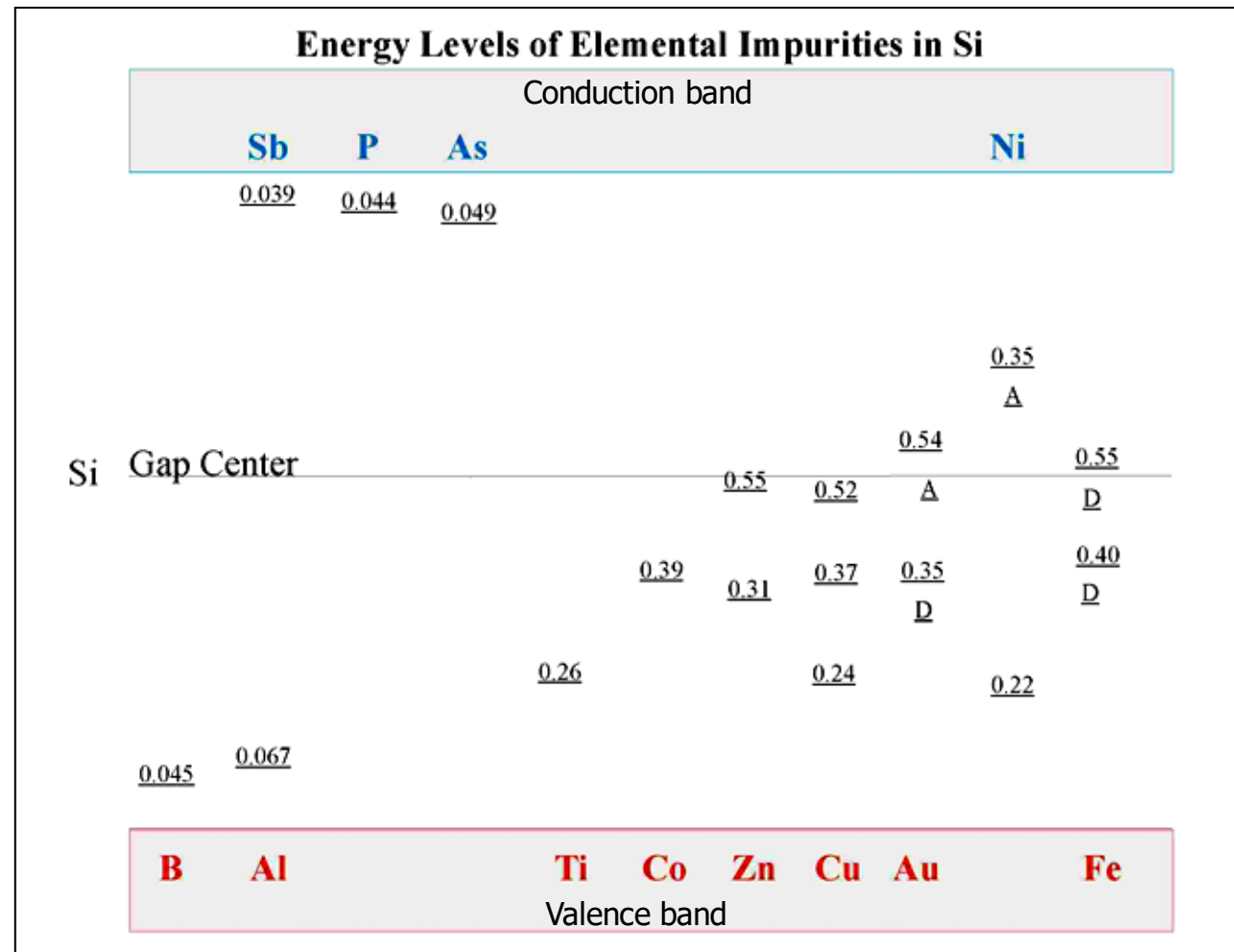
Device **performance**

This may be affected due to the presence of small pieces of contamination, not detected during quality checks.

Device **reliability**

Failure of a device due to the presence of a small amount of metallic contaminants that get into the wafer during processing, and are not detected during device testing. These contaminants can travel inside the device and end up in electrically sensitive areas and cause device failure.

Semiconductor devices: Ionic contaminants



Metals

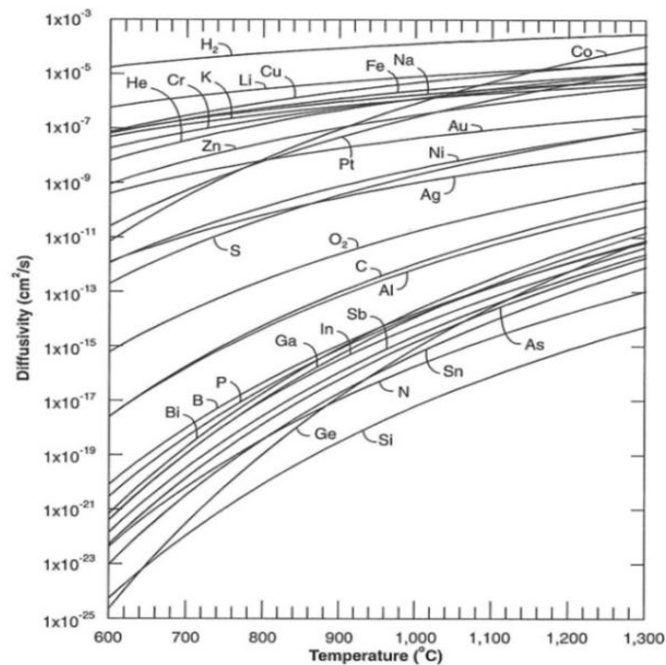
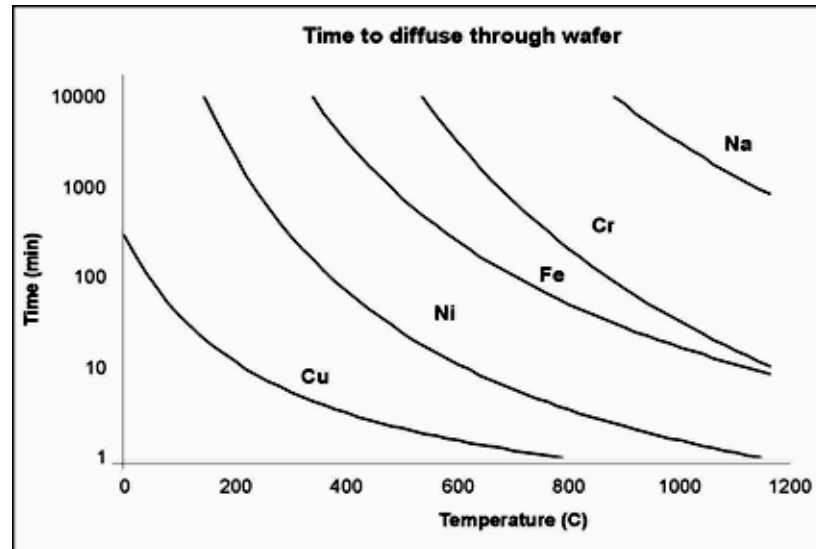
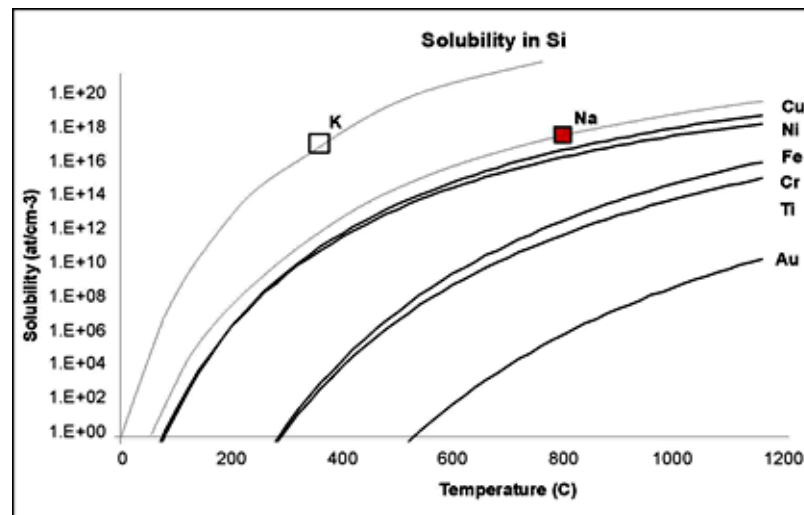


Figure 1.23: Diffusivities of selected impurities in silicon versus temperature



Example: Impurities in Silicon

Very useful impurities!!
'Shallow' impurity levels

p-type dopants: B, Al, Ga, In
n-type dopants: P, As, Sb

Benign impurities.

O₂, C

Potentially dangerous impurities!!
'Deep' impurity levels

Slowly diffusing & rare metals:
La, Y, Zr, Hf, Ta, Mo...

Very dangerous for devices!!
'Deep' impurity levels

Fe, Cu, Ni,
Cr, Mn, Au,...

Effect of contamination

- Fe, Cr, Cu can dissolve in Si (and form silicides).
- Al contamination affects wafer oxidation time.
- Ni can increase Si etching rate.
- Mobile ions like Na^+ , K^+ in SiO_2 result in shifting of V_t .
- Metals can degrade the insulating and capacitive properties of dielectrics.

Effect of contamination

On MEMS devices

- Many of the particle and other contamination issues of concern in IC manufacturing are valid for MEMS processing.
- The effects of contaminants tend to be process and very much design specific.
- Wafer cleaning and surface preparation - very important.

Effect of contamination

On MEMS devices

Technology	Contamination Effects
Surface micromachining	Non uniformity in the films: (alter the mechanical properties, optical and electrical) Effect on static friction (moving parts) 1/4 and 1/5 of the smallest geometry can be a fatal defect
Bulk micromachining (3D structuring) FS/BS processing	Scratches, small notches and residues Bulk etching definition problems FS - IC cleanroom BS - less clean environment
High Aspect Ratio	Particles, small defects - major consequences for etch profile, etch depth control
Wafer bonding	Very sensitive to particles and outgassing

Contamination prevention in EKL



Contamination prevention in EKL

What do we do and why is this important for EKL?

- Bipolar, CMOS and MEMS processing all in the same lab.
- Many different materials are being used (more than 300).
- Equipment and other tools are often used for different processes.

This results in a high risk for cross contamination!

Our aim: Reliable and reproducible process & research results.

EKL's Preventive Cross Contamination strategy

- Full flowchart control.
- Systematic monitoring of contamination.
- Equipment classification (clean, semi clean, contaminated).
- Corrective action plan in case of contamination incidents.
- Consistent lab ware policy.
- Clear equipment policy.

Full flowchart control

It is not allowed to process WITHOUT an APPROVED FLOWCHART.

- A flowchart always needs to be checked and approved by the user's mentor.
- When CMOS incompatible materials are going to be used, approval will also be needed from the EKL contamination officer ([Hande Aydoğmus](#)).

An approved flowchart involves:

- Reading the rules for Prevention of Cross Contamination (PCC) on the EKL [SharePoint](#) website.
- Checking the Process Guide Table (PGT) for the used materials on the [SharePoint](#) website.
- Reading the equipment [manuals](#) & [restrictions](#) to check the possibilities.
- Acquiring a run (= batch) number.

To get a run/batch number you need:

- The first page of your [approved flowchart](#) (mail it to the process manager).
(It should include your name and your mentors name, the materials that you are going to use and which labs that you intend to use.)

Systematic monitoring of contamination

Monitoring:

- Definition of EKL's [contamination threshold level](#).
- Perform regular contamination checks of the equipment.

Impurity threshold levels:

- 10^{11} atoms/cm² for critical contaminants:
Red metals: Ag, Au, Cu, Fe, Sb, Cr, Ni, Pt, Zn...
- 10^{12} atoms/cm² for non-critical contaminants:
Green metals Al, Al(1%Si), Ti, Mo, Zr, ...

Equipment classification

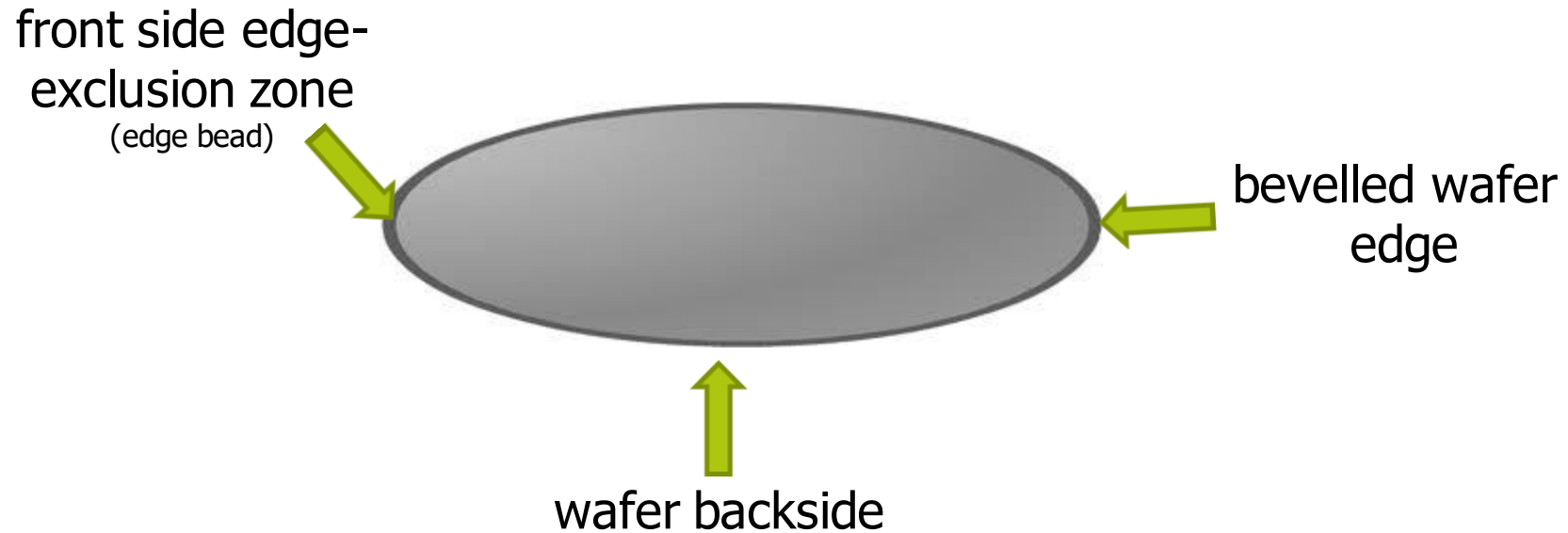
Clean	Semi-clean	Contaminated
Tempress A1 furnace	ASML PAS 5500/80 stepper	AML bonder
Tempress A2 furnace	EVG120 coater/developer	Aixtron Black Magic
ASMI Epsilon 2000	EVG101 spray coater	Tempress A4 furnace
...	Novellus Concept 1	Tempress F3 furnace
...	Plasma etchers (except AMS110)	AMS110 DRIE etcher
...	Trikon Sigma 204	CHA metal evaporator
...	TePla GIGAbatch 360	Tepla plasma 300
...	...	...
...	...	...

No metals allowed in the system

Only green metals allowed
(or protective measures required)

Red metals allowed

Contaminants must be eliminated from:



- Keep the **indicated areas clean** to avoid spreading of contamination between chucks, robot arms, cassettes, etc...
- If necessary, **use a barrier layer** (e.g. oxide, TiN, photoresist) as protection. (Partly) remove this layer (together with possible contamination) in a dedicated etch bath, before going to the next process step.

Corrective action plan in case of contamination incidents

Procedure in case of a **contamination incident**:

User actions:

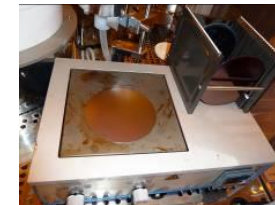
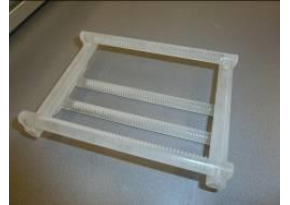
- Stop the process and put the machine down.
(Label the tool with a **STOP** sign and report it down in the NanoLabNL Information system.)
- Inform the staff members that are responsible for the equipment.
(Do **not** call the **Facility Team** for equipment contaminations, because they are not responsible for the tools.)

Equipment owner actions:

- Check if it is possible to clean the machine with the proper defined de-contamination procedure.
- Check if it is necessary to exchange machine parts.
- Always inform the contamination officer.

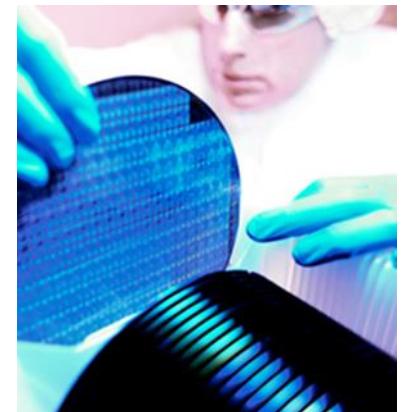
Consistent equipment/lab ware policy

- Check the equipment restrictions and read the manuals.
- Use dedicated holders, chucks, shields or boats.
- Use cleanroom paper or a cleanroom towel on a microscope chuck under contaminated wafers.
- Don't skip cleaning process steps.
- Use transport/carrier/dummy wafers.
- Encapsulation of (red) metals.
- Process chamber cleaning.



Consistent equipment/lab ware policy

- Use a proper coloured clean **process box**, and always put an EKL **label** on it.
 - ⇒ Max. number of coloured boxes allowed per user: **3**.
- Separate your **non-contaminated** tweezers from **contaminated tweezers**.
 - ⇒ Label contaminated tweezers, and don't use them for uncontaminated wafers.
- Regularly **clean** your tweezers. (Ask help from the EKL lab assistants.)
- Keep your tweezers!
 - ⇒ Label them with your name.
- Use proper and clean lab ware. (Ask help from the EKL lab assistants.)
- **Behave properly in the lab.**
 - ⇒ Use appropriate wafer handling tools, NOT your fingers.
 - ⇒ Make no unnecessary movements.



This is the wrong way.

Contamination ethics

- **We all make mistakes**
Take responsibility. Inform EKL staff when you made a mistake, so we can minimize the negative effect of it.
- **Ask questions**
In case of doubts, ask questions. Do not just try something.
- **Don't allow bad habits to propagate**
Be conscientious about your own habits and don't be shy about correcting other users. You have the right to correct people. Offer the correct solution, but respect the person.
- **We are a multi-application laboratory**
Remember that every user is making a different type of device.
- **This is a shared facility**
Everything that you do does not only affect your work, but also that of your fellow lab members.

Conclusions

- Contamination control is a continuous battle.
- User behavior has a critical impact on contamination.
- The proper procedures must be followed at all times.
- Think about your actions! **Be an active participant!**

KEEP THE LAB FREE FROM CONTAMINATION!

